

Colgate Men's Rowing Performance Data

1 Overview

This is some work I did for the Colgate rowing team. This is a good example of what could be a test question, we walk through data cleaning, summary, analysis, and modelling.

```
1 default_chunk_hook <- knitr::knit_hooks$get("chunk")
2
3 latex_font_size <- c(
4   "Huge",
5   "huge",
6   "LARGE",
7   "Large",
8   "large",
9   "normalsize",
10  "small",
11  "footnotesize",
12  "scriptsize",
13  "tiny"
14 )
15
16 knitr::knit_hooks$set(chunk = function(x, options) {
17   x <- default_chunk_hook(x, options)
18   if (options$size %in% latex_font_size) {
19     paste0("\n \\", options$size, "\n\n", x, "\n\n \\", normalsize")
20   } else {
21     x
22   }
23 })
```

2 Charts and stuff

Data cleaning! Remove rowers that quit, and make sure that people that vary their usage of their first and last names are coerced to the same person.

```
1 data = read.csv("/Users/hudson/Downloads/CSV_Colgate_MROW_Updated_3:12:26.csv") |>
2   filter(Name != "Harry" & Name != "David" & Name != "Ethan") |>
3   mutate(Name = case_when(Name == "Williams" ~ "Luca",
4     .default = Name))
```

```
1 data = data |>
2   group_by(Workout.Number) |>
3   mutate('Percent of Mean' = Watts/mean(Watts)) |>
4   ungroup()
```

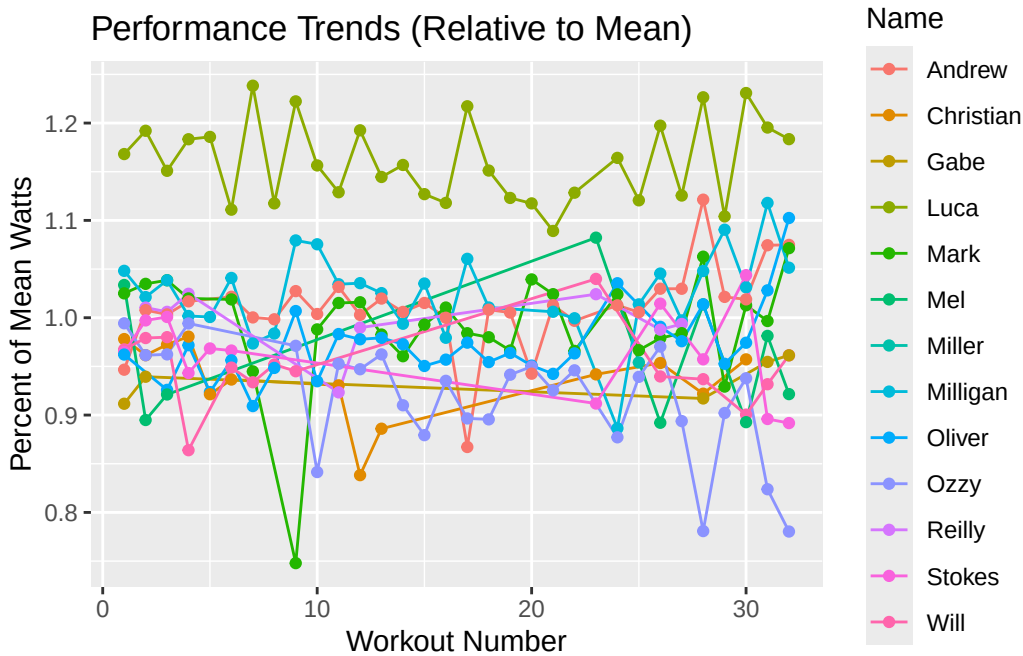
ggplot! Try and interpret this plot. How could I have made it better? (Watts are how we measure power in rowing)

```
1 library(dplyr)
2 library(ggplot2)
```

```

3 library(stringr)
4
5 data |>
6   mutate(Name = str_extract(Name, "[A-Za-z]+")) |>
7   ggplot(aes(x = Workout.Number,
8             y = `Percent of Mean`,
9             color = Name,
10            group = Name)) +
11   geom_line() +
12   geom_point() +
13   labs(
14     x = "Workout Number",
15     y = "Percent of Mean Watts",
16     title = "Performance Trends (Relative to Mean)"
17   )

```



Some more data cleaning! This is converting seconds to watts from the rowers' best races this season.

```

1 data <- data |>
2   mutate(Name_key = str_extract(Name, "[A-Za-z]+"),
3          Best_2k = case_when(
4            Name_key == "Williams" ~ 2.80 * (500 / (365.1 / 4))^3,
5            Name_key == "Milligan" ~ 2.80 * (500 / (373.03 / 4))^3,
6            Name_key == "Andrew" ~ 2.80 * (500 / (375.9 / 4))^3,
7            Name_key == "Oliver" ~ 2.80 * (500 / (379.7 / 4))^3,
8            Name_key == "Mark" ~ 2.80 * (500 / (387.6 / 4))^3,
9            Name_key == "Mel" ~ 2.80 * (500 / (388.2 / 4))^3,
10           Name_key == "Christian" ~ 2.80 * (500 / (390.1 / 4))^3,
11           Name_key == "Gabe" ~ 2.80 * (500 / (393.3 / 4))^3,
12           Name_key == "Will" ~ 2.80 * (500 / (396.4 / 4))^3,
13           Name_key == "Stokes" ~ 2.80 * (500 / (386.0 / 4))^3,
14           Name_key == "Ozzy" ~ 2.80 * (500 / (413.0 / 4))^3,
15           Name_key == "Harry" ~ 2.80 * (500 / (410.7 / 4))^3,
16           .default = NA_real_
17         )) |>

```

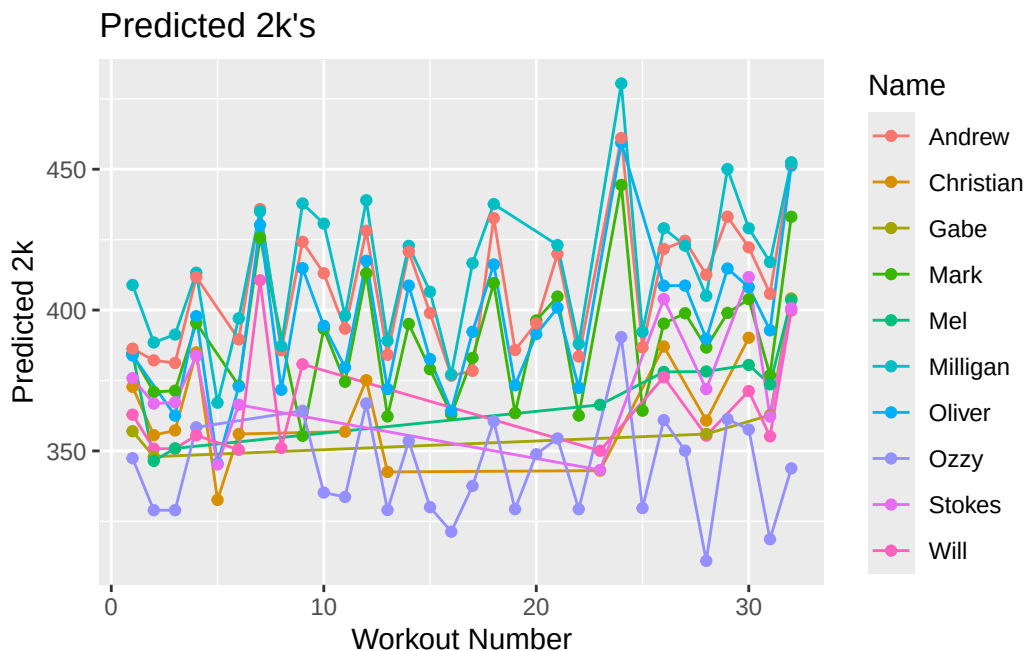
```

18 select(-Name_key)

1 data <- data |>
2   mutate(Predicted.2k.new = (Watts * ((Time..s.)^0.19) * Best_2k)^0.455)

1 library(dplyr)
2 library(ggplot2)
3 library(stringr)
4
5 data |> filter(!is.na(Predicted.2k.new)) |>
6   mutate(Name = str_extract(Name, "[A-Za-z]+")) |>
7   ggplot(aes(x = Workout.Number,
8             y = `Predicted.2k.new`,
9             color = Name,
10            group = Name)) +
11   geom_line() +
12   geom_point() +
13   labs(
14     x = "Workout Number",
15     y = "Predicted 2k",
16     title = "Predicted 2k's"
17   )

```



This predictive model does not include any numerical summary. If you were to include a numerical summary, what would you report?

```

1 data = read.csv("/Users/hudson/Downloads/CSV_Colgate_MROW_Updated_3:12:26.csv")

1 data = data |>
2   group_by(Workout.Number) |>
3   mutate('Percent of Mean' = Watts/mean(Watts)) |>
4   ungroup()

1 data.no.hr <- data |>
2   filter(Time..s. != 3600)

```

```

1 data.no.hr = data.no.hr |>
2   group_by(Name) |>
3   mutate(
4     `2k.Watts` = Watts[Workout.Number == 31][1],
5     Predicted.2k.new = (Watts * ((Time..s.)^0.19) * `2k.Watts`)^0.455
6   ) |>
7   ungroup()

```

```

1 data.no.hr = data.no.hr |>
2   mutate(`Predicted.2k.time.s` = ((2.8 / `Predicted.2k.new`)^(1/3)) * 2000,
3   `Predicted.2k.time.m` = sprintf(
4     "%d:%02d",
5     floor(`Predicted.2k.time.s` / 60),
6     round(`Predicted.2k.time.s` %% 60)
7   )
8 )

```

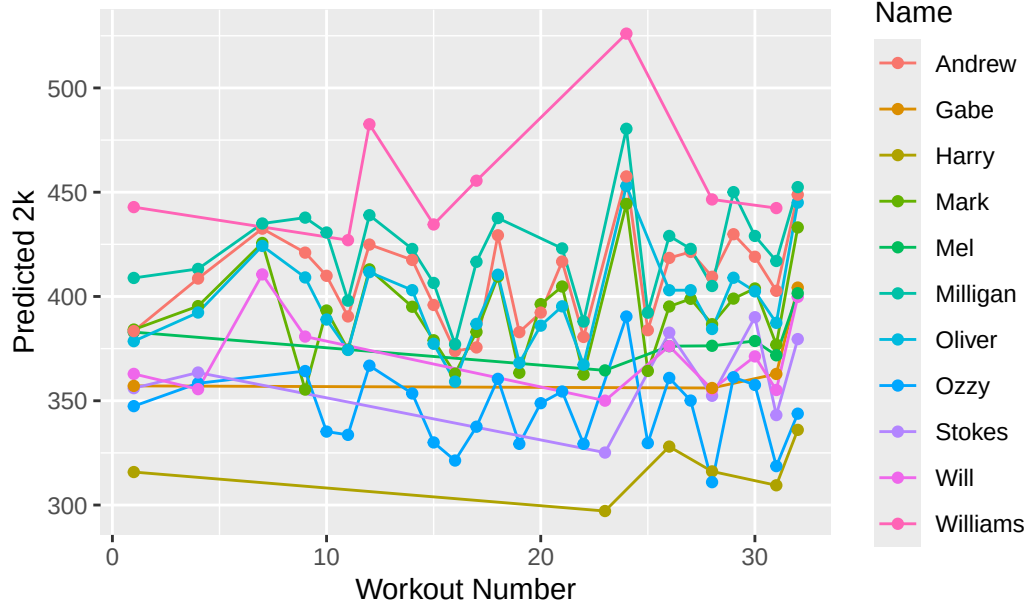
This is a plot of race predictions under a different model! How would you interpret, change, and summarize this plot?

```

1 library(dplyr)
2 library(ggplot2)
3 library(stringr)
4
5 data.no.hr |> filter(!is.na(Predicted.2k.new)) |>
6   mutate(Name = str_extract(Name, "[A-Za-z]+")) |>
7   ggplot(aes(x = Workout.Number,
8             y = `Predicted.2k.new`,
9             color = Name,
10            group = Name)) +
11   geom_line() +
12   geom_point() +
13   labs(
14     x = "Workout Number",
15     y = "Predicted 2k",
16     title = "Predicted 2k's"
17   )

```

Predicted 2k's



```

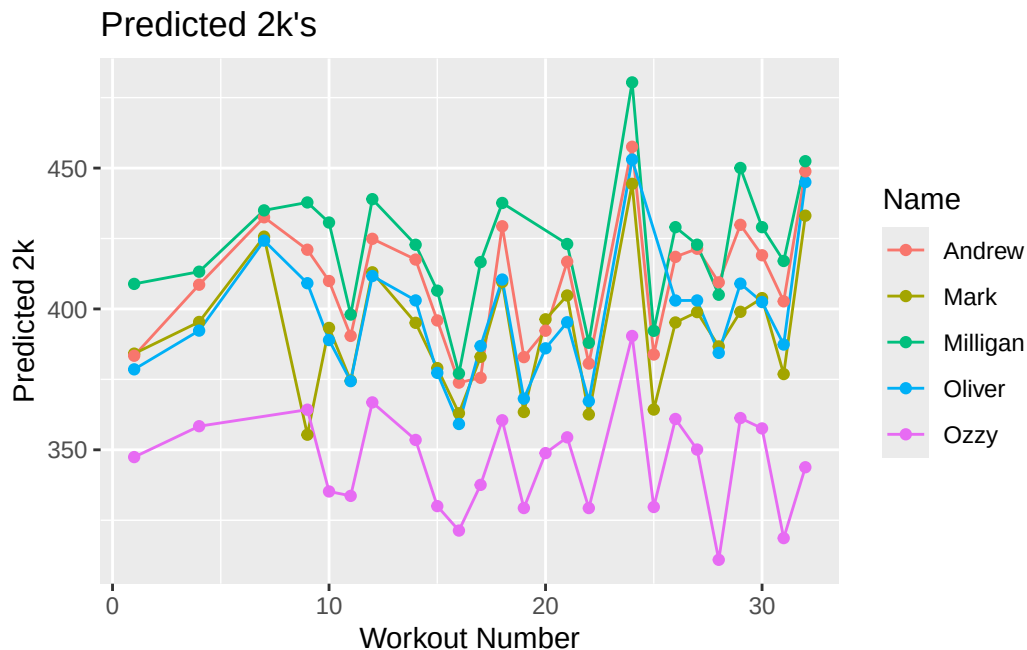
1 # #| echo: false
2 # library(dplyr)
3 # library(ggplot2)
4 # library(stringr)
5
6 # data.no.hr |> filter(!is.na(Predicted.2k.new)) |>
7 #   mutate(Name = str_extract(Name, "[A-Za-z]+")) |>
8 #   ggplot(aes(x = Workout.Number,
9 #             y = `Predicted.2k.time.m`,
10 #             color = Name,
11 #             group = Name)) +
12 #   geom_line() +
13 #   geom_point() +
14 #   labs(
15 #     x = "Workout Number",
16 #     y = "Predicted 2k Time",
17 #     title = "Predicted 2k's (time)"
18 #   )

```

```

1 athletes <- c("Milligan", "Ozzy", "Andrew", "Oliver", "Mark", "Lucca", "Matheson")
2
3
4 data.no.hr |> filter(!is.na(Predicted.2k.new)) |>
5   filter(Name %in% athletes) |>
6   mutate(Name = str_extract(Name, "[A-Za-z]+")) |>
7   ggplot(aes(x = Workout.Number,
8             y = `Predicted.2k.new`,
9             color = Name,
10             group = Name)) +
11   geom_line() +
12   geom_point() +
13   labs(
14     x = "Workout Number",
15     y = "Predicted 2k",
16     title = "Predicted 2k's"

```



This is a small subset of rowers from this year. What would you say about this chart? Is there another chart that could work better here?

```

1 Calibration = data.no.hr|>
2   filter(Workout.Number == 31) |>
3   dplyr::select(`2k.Watts`, `Predicted.2k.new`) |>
4   mutate(delta = `2k.Watts` - `Predicted.2k.new`)
5
6 summary(
7   Calibration,
8   mean = mean()
9 ) |>
10
11 knitr::kable()

```

2k.Watts	Predicted.2k.new	delta
Min. :307.6	Min. :309.5	Min. :-1.881
1st Qu.:352.7	1st Qu.:349.2	1st Qu.: 3.578
Median :378.8	Median :371.8	Median : 6.996
Mean :378.9	Mean :371.6	Mean : 7.272
3rd Qu.:405.8	3rd Qu.:395.0	3rd Qu.:10.750
Max. :461.4	Max. :442.4	Max. :19.016

This is a numerical summary that helps us judge the accuracy of our model. How would you go about this procedure?

```

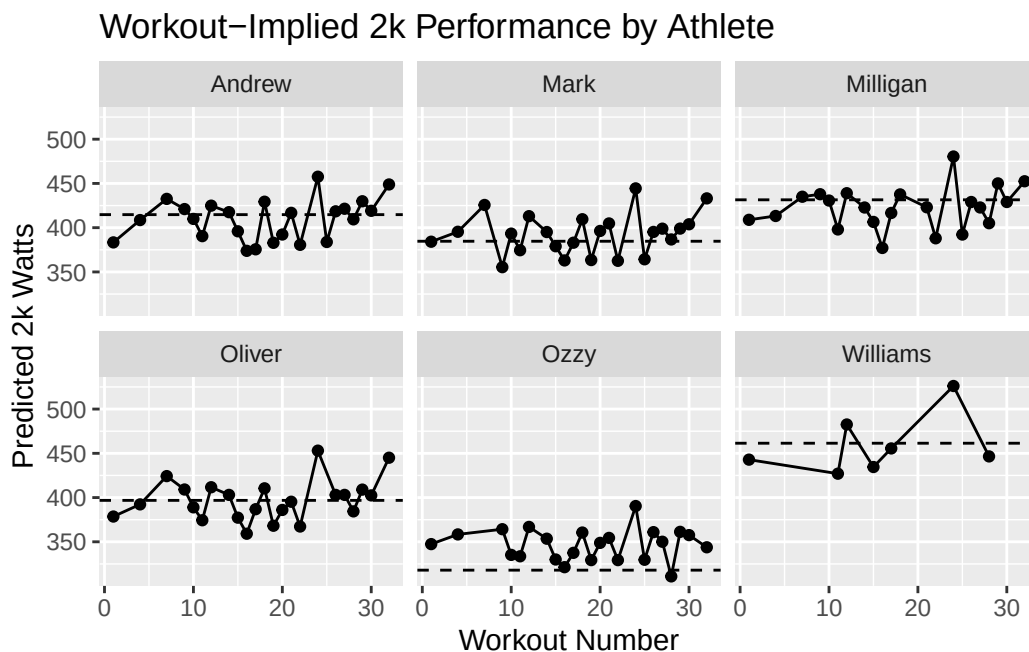
1 library(dplyr)
2 library(ggplot2)
3 library(stringr)
4
5
6 athletes <- c("Mark", "Williams", "Milligan", "Andrew", "Oliver", "Ozzy")
7
8 data.no.hr |>

```

```

9 mutate(Name = str_extract(Name, "[A-Za-z]+")) |>
10 filter(
11   Workout.Number != 31,
12   !is.na(Predicted.2k.new),
13   Name %in% athletes
14 ) |>
15 ggplot(aes(x = Workout.Number, y = Predicted.2k.new)) +
16   geom_point() +
17   geom_line() +
18   geom_hline(aes(yintercept = `2k.Watts`), linetype = "dashed") +
19   facet_wrap(~Name) +
20   labs(
21     x = "Workout Number",
22     y = "Predicted 2k Watts",
23     title = "Workout-Implied 2k Performance by Athlete"
24   )

```



```

1 library(dplyr)
2 library(stringr)
3 library(broom)
4 library(knitr)
5
6 athletes <- c("Mark", "Williams", "Milligan", "Andrew", "Oliver", "Ozzy")
7
8 table_out <- data.no.hr |>
9   mutate(Name = str_extract(Name, "[A-Za-z]+")) |>
10  filter(Name %in% athletes, !is.na(Predicted.2k.new)) |>
11  group_by(Name) |>
12  do({
13    m <- lm(Predicted.2k.time.s ~ Workout.Number, data = .)
14    tibble(
15      Intercept = coef(m)[1],
16      Slope = coef(m)[2],
17      R2 = summary(m)$r.squared,
18      p_value = summary(m)$coefficients[2,4]

```

```

19   )
20   }) |>
21   ungroup()
22
23   kable(
24     table_out,
25     digits = 3,
26     col.names = c("Athlete", "Intercept (sec)", "Slope (sec/workout)", "R2", "p-value"),
27     caption = "Linear Model of Predicted 2k Time vs Workout Number"
28   )

```

Table 2: Linear Model of Predicted 2k Time vs Workout Number

Athlete	Intercept (sec)	Slope (sec/workout)	R ²	p-value
Andrew	383.652	-0.196	0.061	0.234
Mark	387.961	-0.135	0.026	0.441
Milligan	378.399	-0.130	0.030	0.426
Oliver	387.439	-0.191	0.055	0.270
Ozzy	400.004	0.101	0.015	0.573
Williams	369.365	-0.179	0.047	0.608

This is a little bit outside the scope of our course. If you wanted to answer the question: are athletes getting faster from the start of the season to the end, what would you do? (hint: hypothesis testing)

```

1   library(dplyr)
2
3   # attach true 2k watts (Workout 31) to each athlete
4   data.model <- data.no.hr |>
5     group_by(Name) |>
6     mutate(`2k.Watts` = Watts[Workout.Number == 31][1]) |>
7     ungroup()
8
9   # fit global fatigue exponent
10  fit <- nls(
11    `2k.Watts` ~ Watts * (Time..s. / 390)^d,
12    data = data.model |> filter(Workout.Number != 31),
13    start = list(d = 0.07)
14  )
15
16  # extract fitted exponent
17  d_hat <- coef(fit)["d"]
18
19  # compute predicted 2k watts for every workout
20  data.model <- data.model |>
21    mutate(
22      Predicted.2k = Watts * (Time..s. / 390)^d_hat
23    )
24
25  # optional: check prediction stability by athlete
26  sd_table <- data.model |>
27    group_by(Name) |>
28    summarize(
29      sd_pred = sd(Predicted.2k.new, na.rm = TRUE),
30      mean_pred = mean(Predicted.2k.new, na.rm = TRUE)
31    )

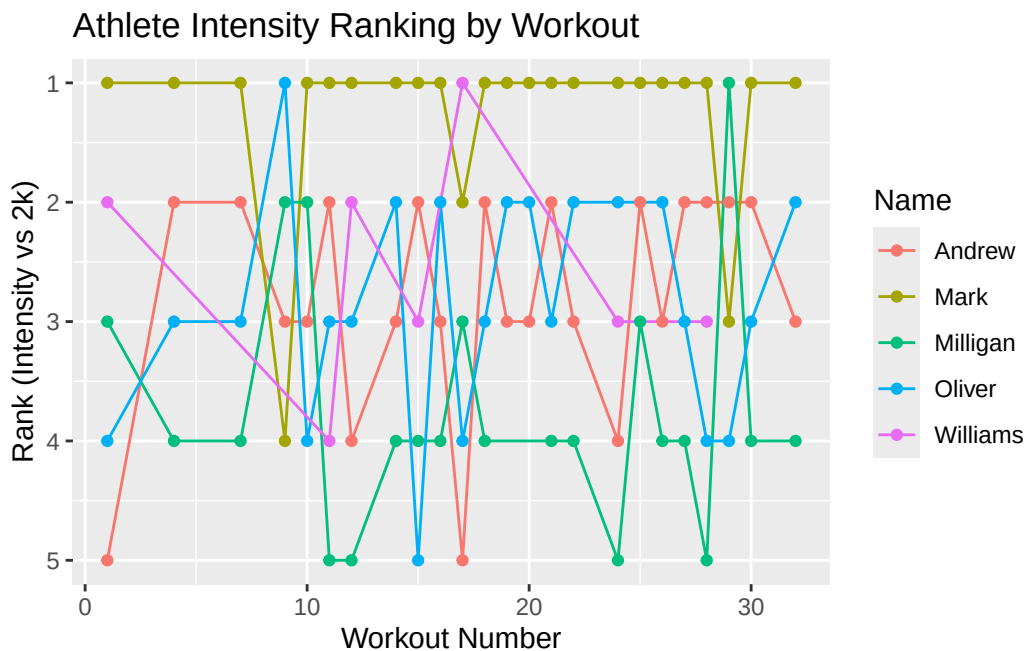
```



```

1 athletes <- c("Mark", "Williams", "Milligan", "Andrew", "Oliver")
2
3 data.rank <- data.model |>
4   mutate(Relative.Intensity = Watts / `2k.Watts`) |>
5   filter(Name %in% athletes) |>
6   group_by(Workout.Number) |>
7   mutate(Rank = rank(-Relative.Intensity)) |>
8   ungroup()
9
10 library(ggplot2)
11
12 data.rank |>
13   filter(Workout.Number != 31) |>
14   ggplot(aes(x = Workout.Number, y = Rank, color = Name, group = Name)) +
15     geom_line() +
16     geom_point() +
17     scale_y_reverse() +
18     labs(
19       x = "Workout Number",
20       y = "Rank (Intensity vs 2k)",
21       title = "Athlete Intensity Ranking by Workout"
22     )

```



This graphic was the one that my team and coaching staff looked at the most. Why would I include it? What actionable insights can be derived from it? How would you change it, or add to it?